BOOKSTORE DATABASE ANALYSIS USING SQL

Online Book Store



Presented by: Prem Kumar

Project Type: SQL Data Analysis Project

Tools Used: PostgreSQL and pgAdmin

Objective: To analyze bookstore data and solve real-world business questions using SQL.



About Me

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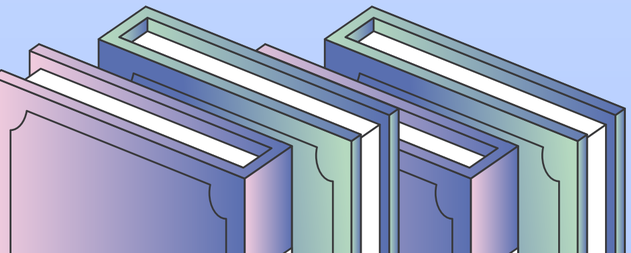
Hello everyone,

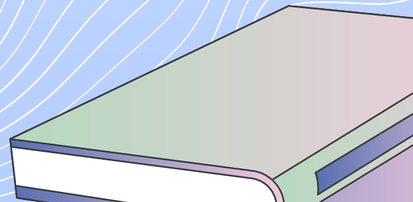
My name is Prem Kumar. I am currently pursuing a Bachelor of Computer Applications (BCA), 3rd Semester.

I am interested in Data Analytics and enjoy working with data to solve problems and generate meaningful insights.

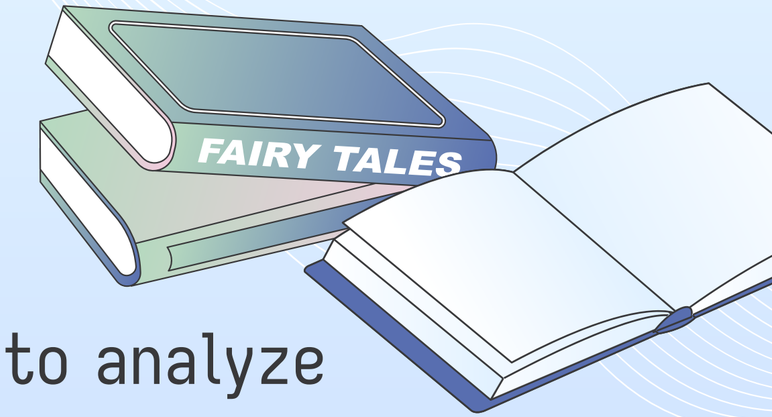
I have learned SQL, Excel, Power BI, and basic Python as part of my data analytics learning journey.

This project helped me improve my SQL querying and database analysis skills.





Project Introduction



The Bookstore Database Analysis project is designed to analyze book, customer, and order data using SQL.

In this project, I created database tables, imported CSV files, and solved basic to advanced SQL questions.

The main goal was to understand how SQL can be used to manage data, retrieve information, and answer real-world business questions.

Dataset Description

I used three CSV files for this project:



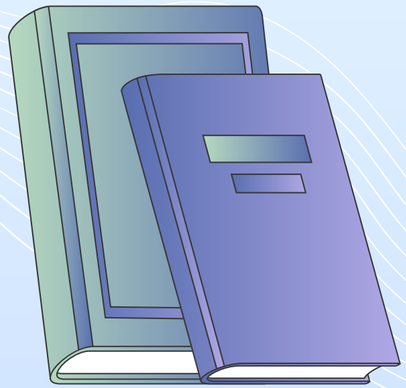
Books.CSV



Customer.CSV

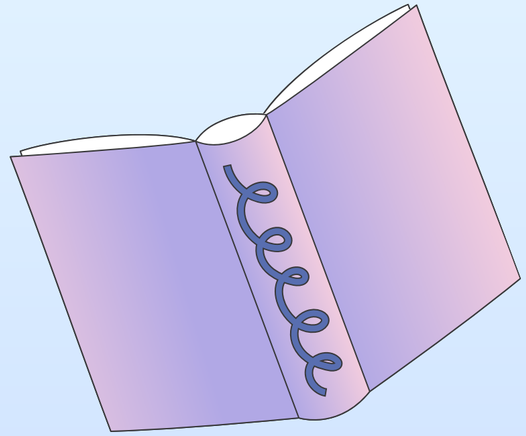


Orders/CSV





Project Workflow



Created database tables using SQL.

1. Defined primary keys and foreign key relationships.
2. Imported data from three CSV files.
3. Checked the imported data.
4. Wrote SQL queries to answer business questions.
5. Analyzed the results to extract useful information.

This workflow helped me understand the complete process of database creation and SQL-based data analysis.

SQL Concepts Used

During this project, I worked with the following SQL concepts:

- SELECT: To retrieve data from tables.
- WHERE: To filter records based on conditions.
- JOIN: To combine data from multiple tables.
- GROUP BY: To group records for analysis.
- HAVING: To filter grouped results.
- Aggregate Functions: SUM(), AVG(), COUNT(), MIN(), and MAX() to perform calculations and summarize data.
- ORDER BY: To sort query results.
- Primary Keys and Foreign Keys: To define relationships between tables.

create tables

```sql
drop table if exists books;

create table books (
    book_id int primary key,
    tital varchar(100),
    Author VARCHAR(100),
    genre VARCHAR(50),
    published_year INT,
    price numeric(10,2),
    stock int

);
```

```sql
drop table if exists orders;
create table orders (
    order_id int primary key,
    customer_id int,
    book_id int,
    order_date date,
    quantity int,
    total_Amount int
);
```

```sql
drop table if exists customers;

create table customers (
    customer_id INT primary key,
    name VARCHAR(50),
    email VARCHAR(150),
    phone bigint,
    city VARCHAR(50),
    country VARCHAR(50)
);
```

Retrieve all books in the "Fiction" genre;

```sql
SELECT *
FROM BOOKS
WHERE
GENRE = 'Fiction'
```

book_id [PK] integer	tital character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	The Alchemist	Paulo Coelho	Fiction	1988	299.00	12

find books published after the year 1950:

```sql
SELECT *
FROM BOOKS
WHERE PUBLISHED_YEAR > 1950
```

book_id [PK] integer	tital character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	The Alchemist	Paulo Coelho	Fiction	1988	299.00	12
2	Wings of Fire	A. P. J. Abdul Kalam	Autobiography	1999	250.00	8
3	Atomic Habits	James Clear	Self-help	2018	450.00	15
4	Ikigai	Hector Garcia	Self-help	2016	325.00	10
6	Rich Dad Poor Dad	Robert Kiyosaki	Finance	1997	350.00	9
7	Harry Potter	J. K. Rowling	Fantasy	1997	499.00	5
10	Deep Work	Cal Newport	Productivity	2016	375.00	11

List all customers from the canada

```sql
SELECT
    *
FROM
    CUSTOMERS
WHERE
    COUNTRY = 'canada'
```

customer_id [PK] integer	name character varying (50)	email character varying (150)	phone bigint	city character varying (50)	country character varying (50)

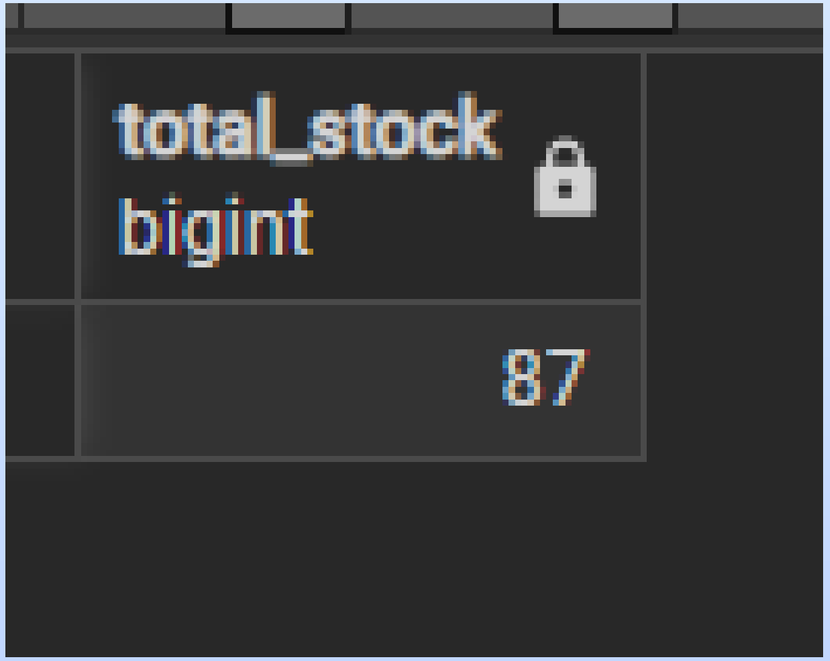
show orders placed in november 2023

```sql
SELECT
    *
FROM
    ORDERS
WHERE
    ORDER_DATE BETWEEN '01-09-2023' AND '30-09-2023'
```

order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount integer

retrieve the total stock of books available:

```sql
SELECT
    SUM(STOCK) AS TOTAL_STOCK
FROM
    BOOKS
```



Find the details of the most expensive book

```sql
SELECT
    *
FROM
    BOOKS
ORDER BY
    PRICE DESC
LIMIT
    1
```

book_id [PK] integer	tital character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
7	Harry Potter	J. K. Rowling	Fantasy	1997	499.00	5

show all customers who ordered more then 1 quantity of a book

```sql
select c.name, o.quantity
from customers c
join orders o
on o.customer_id = c.customer_id
where o.quantity > 1
```

name character varying (50)	quantity integer
Aarav Sharma	2
Rahul Patel	2
Arjun Mehta	3
Ananya Gupta	2
Neha Joshi	2
Kavya Nair	2

Retrieve all orders where the total amount exceeds 20

```sql
select * from orders where total_amount > 20
```

order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount integer
1	1	3	2026-01-05	2	900
2	2	1	2026-01-08	1	299
3	3	6	2026-01-12	2	700
4	4	2	2026-01-15	1	250
5	5	5	2026-01-18	3	1197
6	6	4	2026-01-22	2	650
7	7	7	2026-01-25	1	499
8	8	10	2026-02-02	2	750
9	9	8	2026-02-06	1	275
10	10	9	2026-02-10	2	440

List all genres available in the books tables

```
select distinct genre from books
```

	genre character varying (50)
1	Finance
2	Self-help
3	Productivity
4	Fantasy
5	Classic
6	Autobiography
7	Fiction

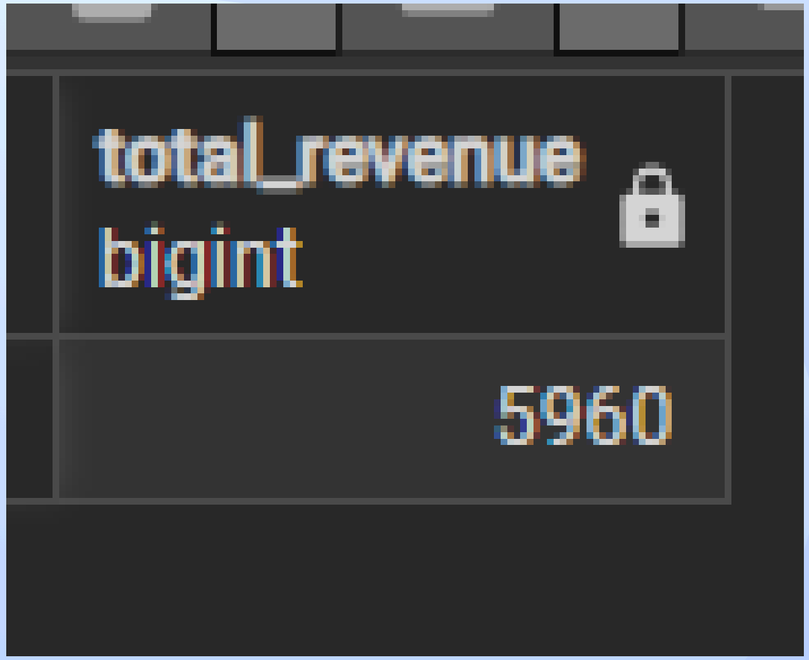
find the book with the lowest stock

```sql
select * from books order by price asc limit 1
```

book_id [PK] integer	tital character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
9	Pride and Prejudice	Jane Austen	Classic	1813	220.00	4

Calculate the total revenue generated from all orders

```sql
select sum(total_amount) as total_revenue from orders
```



Advanced Queries

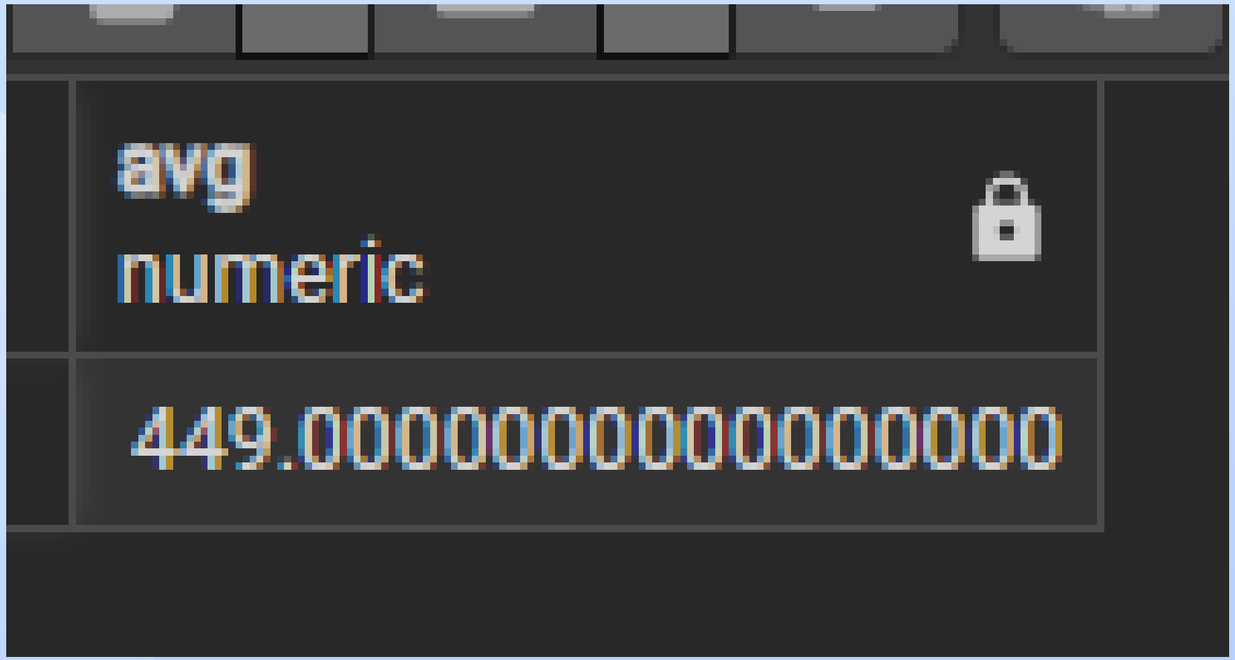
retrieve the total numbers of books sold for each genre:

```sql
select b.genre, sum(o.quantity) as total_numbers_of_books
from books b
join orders o
on o.book_id = b.book_id
group by genre
```

genre character varying (50)	total_numbers_of_books bigint
Finance	3
Self-help	4
Productivity	2
Fantasy	4
Classic	2
Autobiography	1
Fiction	1

find the average price of books in the "Fantasy"

```sql
select avg(price) from books where genre = 'Fantasy'
```



List customers who have placed at least 2 orders:

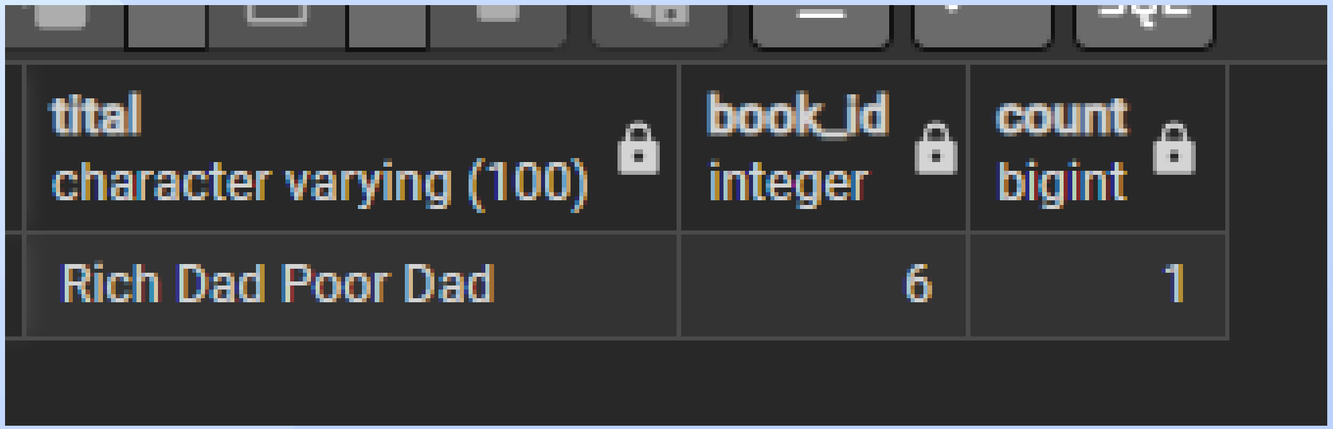
```sql
select customer_id, order_count
from
(select customer_id, count(order_id) as order_count
from orders
group by customer_id) as a  where order_count > 1

-- or

select customer_id, count(order_id)
from orders
group by customer_id
having count(order_id) >= 2
```

find the most frequently ordered book

```sql
select b.tital, o.book_id, count(o.order_id)
from orders o
join books b
on b.book_id = o.book_id
group by b.tital, o.book_id
order by count(order_id) desc limit 1
```



tital character varying (100)	book_id integer	count bigint
Rich Dad Poor Dad	6	1

show the top 3 most expensive books of 'fantasy genre'

```sql
select * from books
where genre = 'Fantasy'
order by price desc limit 3
```

book_id [PK] integer	tital character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
7	Harry Potter	J. K. Rowling	Fantasy	1997	499.00	5
5	The Hobbit	J. R. R. Tolkien	Fantasy	1937	399.00	6

retrieve the total quantity of book sold by each author

```sql
select b.author, sum(o.quantity) as total_quantity
from books b
join orders o
on b.book_id = o.book_id
group by b.author
```

author character varying (100)	total_quantity bigint
J. K. Rowling	1
James Clear	2
Paulo Coelho	1
Cal Newport	2
Jane Austen	2
Napoleon Hill	1
Hector Garcia	2
Robert Kiyosaki	2
J. R. R. Tolkien	3
A. P. J. Abdul Kalam	1

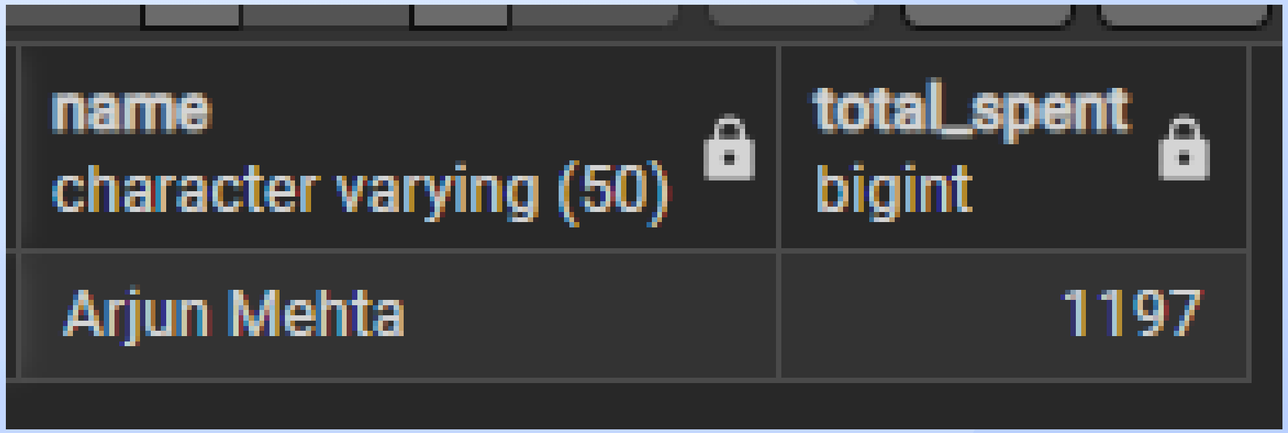
List the cities where customers who spent over 30 are located

```sql
select distinct c.city
from customers c
join orders o
on o.customer_id = c.customer_id
where o.total_amount > 30
```

	city character varying (50)
	Jaipur
	Delhi
	Bengaluru
	Chennai
	Mumbai
	Pune
	Raipur
	Lucknow
	Kolkata
	Ahmedabad

find the customers who spent the most on orders

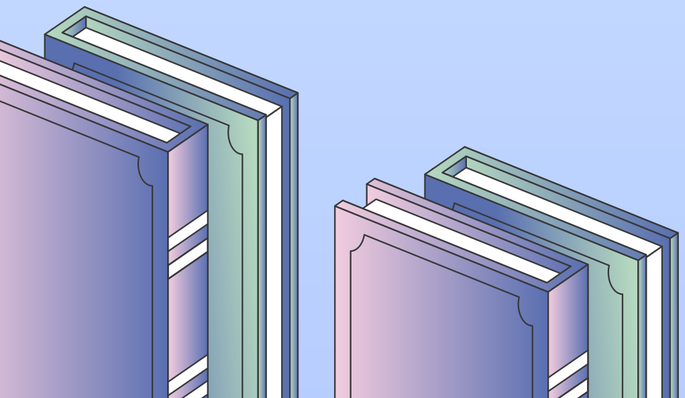
```sql
select c.name, sum(o.total_amount) as total_spent from orders o
join customers c
on o.customer_id = c.customer_id
group by c.name
order by total_spent desc limit 1
```

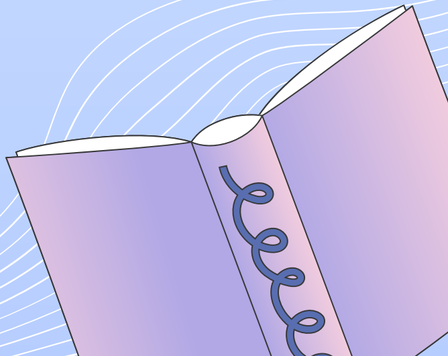


name character varying (50)	total_spent bigint
Arjun Mehta	1197

Conclusion

his project gave me practical experience in using SQL for database management and data analysis.
I learned how to transform raw data into useful information by writing SQL queries and applying different SQL concepts.
This project has strengthened my foundation in SQL and motivated me to build more advanced data analytics projects.
Thank You!
Presented by: Prem Kumar





THANK YOU



